

ML Lab Week 14

CNN Image Classification

Name: Nandani sonale

SRN: PES2UG23CS364

Sec: f

Introduction:

The goal of this lab is to design, build, and train a Convolutional Neural Network (CNN) using PyTorch. This model must be able to accurately classify images of hand gestures into one of three categories: 'rock', 'paper', or 'scissors'.

Model Architecture:

The model consists of three convolutional blocks:

- First block: Conv2d layer with 3 input channels and 16 output channels; kernel size 3×3; padding 1; followed by ReLU activation and 2×2 Max Pooling.
- Second block: Conv2d layer with 16 input channels and 32 output channels; kernel size 3×3; padding 1; followed by ReLU activation and 2×2 Max Pooling.
- Third block: Conv2d layer with 32 input channels and 64 output channels; kernel size 3×3; padding 1; followed by ReLU activation and 2×2 Max Pooling.

Key parameters:

- Kernel size: 3×3 for all convolution layers
- Channels: 3→16→32→64 across Conv2d blocks
- Max Pooling: 2×2 after every convolution block to reduce spatial dimensions

Fully-connected classifier:

- The features after convolutional layers are flattened.
- First linear layer maps the flattened features (64×16×16) to 256 units, applies ReLU activation, and includes a Dropout layer with p=0.3 for regularization.
- Final linear layer maps 256 units to 3 output classes (rock, paper, scissors).

Training and Performance:

- Optimizer: Adam
- Loss Function: CrossEntropyLoss
- Learning Rate: 0.001
- Number of Epochs: 10
- Final Test Accuracy: 97.26%

Conclusion and Analysis:

The model performed very well, achieving high accuracy in classifying rock, paper, and scissors images. This demonstrates that the CNN architecture was effective for the task.

Challenges included correctly setting up image preprocessing and managing the train-test split to ensure fair evaluation.

To further improve accuracy, you could use additional data augmentation techniques to make the model more robust, or experiment with deeper network architectures and fine-tune hyperparameters such as learning rate and dropout rate.